



# Control Flows



Why do we need control flow?

# Every Useful Program Needs to Make Decisions

 A traffic light checks conditions: **red** → stop, **green** → go, **yellow** → slow down. Your Python code does the same thing!

## Game

*Is health > 0?*

→ **Keep playing**

→ **Game over**

## Banking

*Is balance enough?*

→ **Allow transfer**

→ **Decline**

## Weather App

*Is temp < 10°C?*

→ **Show coat icon**

→ **Show sun icon**



# What is control flow?

# What is control flow?

Control flow structures allow you to dictate the order in which **statements** execute in your program.

Python manages control flow using 2 core structures

1

## Conditional Statements

if / elif / else — branch based on True/False checks.

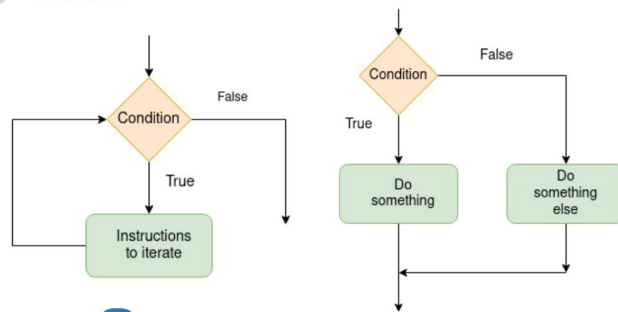
2

## Loop & Transfer Statements

Repeat code and control iteration (covered in a later lesson).



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Python Control flow

# Conditional Statements

# Control Flows: The `if` Statement

Control flow statements let us control the order code runs in, based on conditions.

- **if** keyword — starts the decision check
- **condition** — an expression that is True or False
- **:** colon — REQUIRED at the end of the line
- **indented block** — only runs if condition is True

## Code Sample

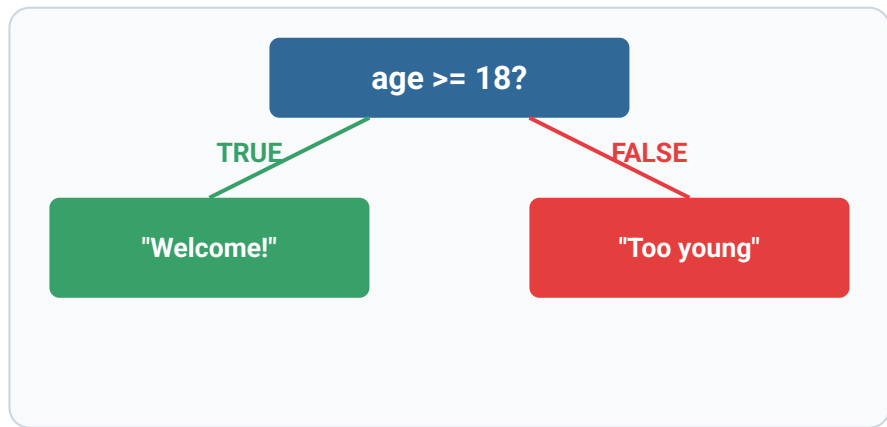


```
# Variable assignment
age = 18

if age ≥ 18:
    print("You can vote!")
```

# Control flow: `else` Statement - A Backup Plan

Program to check age:



Code visualizer:

- <https://programiz.pro/code-visualizer/python>
- <https://pythontutor.com/visualize.html#mode=display>
- <https://learn-code-with-vengleab.vercel.app/>

## Code Sample

```
# Current status
age = 15

if age >= 18:
    print("Welcome!")
else:
    print("Sorry, too young")
```

execution order

L1

| code | step 1/4 | running line 1            |
|------|----------|---------------------------|
| 1    | >        | age = 15                  |
| 2    |          |                           |
| 3    |          | if age >= 18:             |
| 4    |          | print("You can vote!")    |
| 5    |          | else:                     |
| 6    |          | print("Sorry, too young") |

variables (changed in green)

(none yet)

printed output

Challenge: change age to 20 and re-run — what happens to the output?



# Control flow: `elif` Handles Multiple Scenarios

## Grading app:

score >= 90 ? → A — Excellent

score >= 80 ? → B — Good

score >= 70 ? → C — Pass

else (catch-all) → F — Fail

 **Insight:** Python checks conditions top-to-bottom and stops at the FIRST match.

## Code Sample

```
score = 75

if score ≥ 90:
    grade = "A"
elif score ≥ 80:
    grade = "B"
elif score ≥ 70:
    grade = "C"
else:
    grade = "F"

print(f"Grade: {grade}")
```

# Visualize code execution

execution order

L1

code    step 1/7    running line 1

```
1 > score = 75
2
3 if score >= 90:
4     grade = "A"
5 elif score >= 80:
6     grade = "B"
7 elif score >= 70:
8     grade = "C"
9 else:
10    grade = "F"
11
12 print(f"Grade: {grade}")
```

variables    (changed in green)

(none yet)

printed output



Test your knowledge

## What Does This Code Print?



```
x = 7
```

```
if x > 10:  
    print("Big")  
elif x > 5:  
    print("Medium")  
else:  
    print("Small")
```

A) Big

B) Medium

C) Small

D) Nothing – syntax error

## What Does This Code Print?

```
x = 7

if x > 10:
    print("Big")
elif x > 5:
    print("Medium")
else:
    print("Small")
```

A) Big

B) Medium

C) Small

D) Nothing – syntax error

 **Insight:** Answer: B) Medium – x is 7, which is > 5 but NOT > 10, so elif matches first.

# Nested if Statements

# `Nested if` Statements Go Deeper Into Decisions

INDENTATION = NESTING LEVEL

Level 0 — outer if

Level 1 — inner if

Level 2 — inner else

 **Insight:** avoid nesting more than 2–3 levels deep. Consider combining conditions with and / or instead.

## Code Example for login check

execution order

L1

code step 1/6 running line 1

```
1 > logged_in = True
2 is_admin = False
3
4 if logged_in:
5     if is_admin:
6         print("Admin panel")
7     else:
8         print("User dashboard")
9 else:
10    print("Please log in")
11
12 # Output: User dashboard
```

variables (changed in green)

(none yet)

printed output

# Ternary Expressions



# Ternary Expressions: Simple If-else Into One Line

😓 Before (4 lines)

```
if age ≥ 18:  
    label = "Adult"  
else:  
    label = "Minor"
```

😎 After (1 line)

```
label = "Adult" if age ≥ 18 else "Minor"
```



## ANATOMY OF Ternary Expressions

**value\_if\_true** if **condition** else **value\_if\_false**

● **Insight:** use ternary for simple, readable assignments. For complex logic, stick with full if/elif/else.

# Practices

# Positive Number & Even Number

TRY IT YOURSELF · LEVEL 1 · EXERCISES 1-2

## 1 Positive Number

```
ex01.py

number = float(input("Enter a number: "))

# TODO: check if number is positive
```

- Ask for a **number**
- If **positive**, print `Positive`

## 2 Even Number

```
ex02.py

num = int(input("Enter an integer: "))

# TODO: use % to check even
```

- Input an **integer**
- If **even**, print `Even`

# Adult Check & Pass or Fail

TRY IT YOURSELF · LEVEL 1 · EXERCISES 3–4

3

## Adult Check

```
ex03.py

age = int(input("Enter age: "))

# TODO: compare age to 18
```

- Input `age`
- 18+ -> `Adult`, else `Minor`

4

## Pass or Fail

```
ex04.py

score = float(input("Enter score: "))

# TODO: compare score to 50
```

- Input `score`
- $\geq 50$  -> `Pass`, else `Fail`

# Bigger Number & Temperature

TRY IT YOURSELF · LEVEL 1 · EXERCISES 5-6

## 5 Bigger Number

```
ex05.py

a = int(input("First number: "))
b = int(input("Second number: "))

# TODO: compare a and b
```

- Input **two** integers
- Print the **larger** one

## 6 Temperature

```
ex06.py

temp = float(input("Enter temperature: "))

# TODO: compare temp to 30
```

- Input **temperature**
- > 30 -> `Hot`, else `Cool`

# Divisible by 5 & Username Check

TRY IT YOURSELF · LEVEL 1 · EXERCISES 7-8

7

## Divisible by 5

```
ex07.py

num = int(input("Enter a number: "))

# TODO: use % 5
```

- Input a number
- Divisible by 5 -> `Yes`, else `No`

8

## Username Check

```
ex08.py

username = input("Enter username: ")

# TODO: compare to "admin"
```

- Input a **username**
- `admin` -> `Welcome Admin`, else `Access Denied`

# Ternary: Maximum & Ternary: Even/Odd

TRY IT YOURSELF · LEVEL 1 · EXERCISES 9–10

## 9 Ternary: Maximum

```
ex09.py

a = int(input("First: "))
b = int(input("Second: "))

# TODO: one-line ternary
result = ... # a if ... else b
```

- Use **only** a ternary operator
- Print the **larger** number

## 10 Ternary: Even/Odd

```
ex10.py

num = int(input("Enter integer: "))

# TODO: ternary -> "Even" or "Odd"
```

- Use a **ternary** operator
- Print `Even` or `Odd`